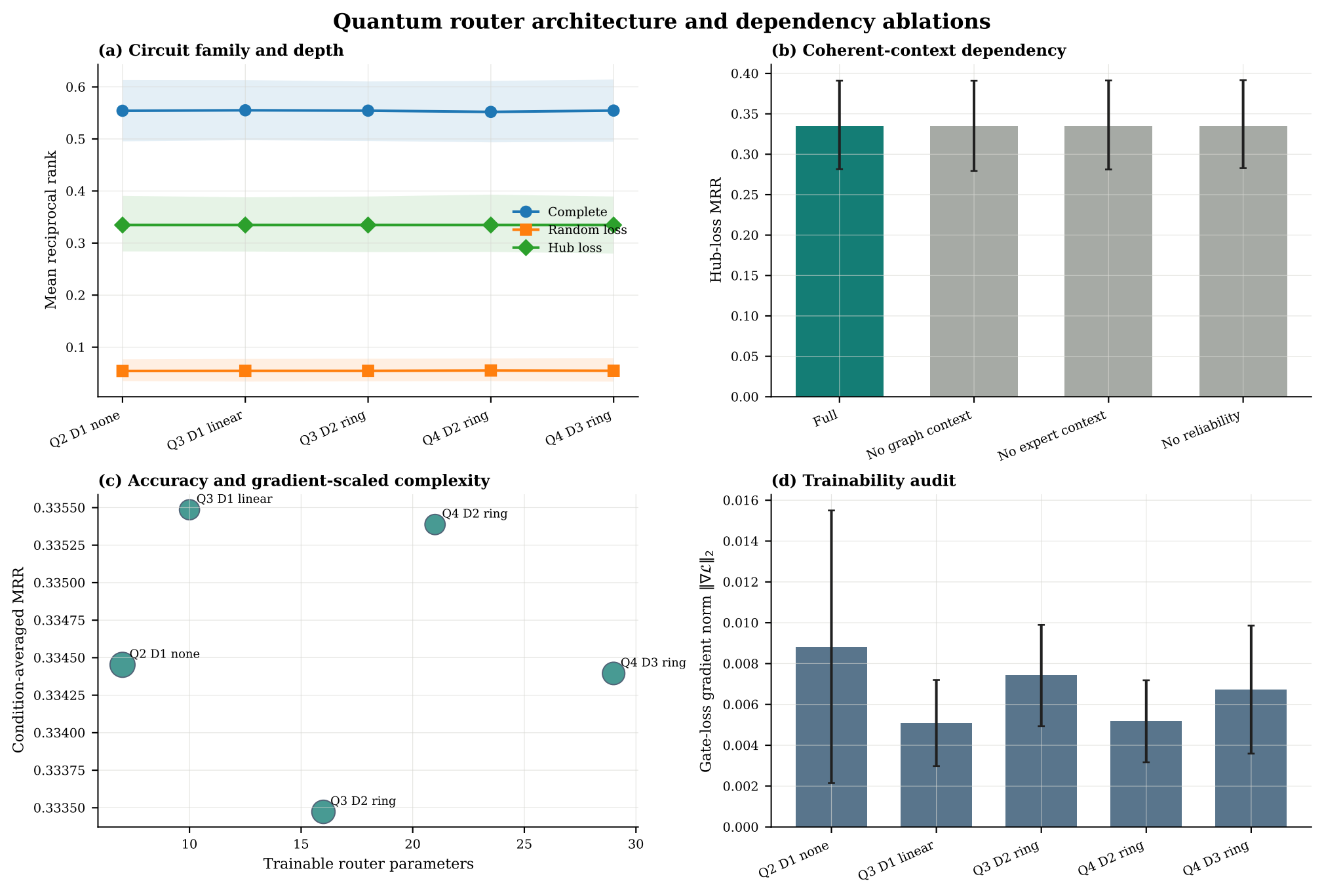
Quantum router architecture and dependency ablations
(a) Circuit family and depth
Mean reciprocal rank
Complete
Random loss
Hub loss
Q2 D1 none
Q3 D1 linear
Q3 D2 ring
Q4 D2 ring
Q4 D3 ring
(b) Coherent-context dependency
Hub-loss MRR
Full
No graph context
No expert context
No reliability
(c) Accuracy and gradient-scaled complexity
Condition-averaged MRR
Trainable router parameters
Q3 D1 linear
Q4 D2 ring
Q2 D1 none
Q4 D3 ring
Q3 D2 ring
(d) Trainability audit
Gate-loss gradient norm $\|\nabla\mathcal{L}\|_2$
Q2 D1 none
Q3 D1 linear
Q3 D2 ring
Q4 D2 ring
Q4 D3 ring